START SMARTER

What Is AI?

You need to start your next history project, but you haven't picked a topic.

Ask your desk, "Give me 10 ideas for my next history project."

Nothing. It's a desk.

Ask a friend, and they can help you brainstorm. Ask AI, and it can give you a list in seconds.

Ask the Desk. Ask AI.



AI is software built to do things that used to take a human brain.

AI is software built to do things that used to take a human brain. Understand a question. Summarize information. Translate a conversation. Suggest ideas for a history project.

That doesn't mean it thinks like your friend, or that all 10 ideas will be good. But you now have a tool that can help with tasks that once needed a person.

Two Types of AI



Recommendation AI

THE JOB

Choose from what already exists.

HOW IT WORKS

Rank the available options and select the likely best match.

EVERYDAY EXAMPLES

Your next Netflix show, Spotify song, or Maps route.



Generative AI

THE JOB

Make something that didn't exist.

HOW IT WORKS

Creates something new from your prompt (the question or instructions you give it).

EVERYDAY EXAMPLES

An email, essay, image, website, song, or video.

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AI IS EVERYWHERE

AI was already part of your day before you started chatting with it.

Where AI Already Lives

Recommends	Navigation	Face Recognition	Voice Assistants	Chatbots
Predicts what you might like next.	Predicts traffic and arrival time.	Checks whether a face matches you.	Turns your speech into words.	Carries on a conversation with you.
YOU'VE SEEN IT IN	YOU'VE SEEN IT IN	YOU'VE SEEN IT IN	YOU'VE SEEN IT IN	YOU'VE SEEN IT IN
Spotify · Netflix TikTok	Google Maps Waze	Phone unlock Photo tagging	Siri · Alexa Hey Google	ChatGPT · Claude Gemini

AI was already part of your day before chatbots arrived.

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This course focuses on generative AI.

You've got a tool that can help.

What can it help you do?

START SMARTER

Why Learn AI?

Pretend you're a scribe in the 1450s. All day, you hand-copy the king's proclamations and send them out across the kingdom. Good work. Steady job. Then one morning, you hear that a guy in Mainz built a machine that prints pages a thousand times faster than you can write them. Two choices: pretend it isn't happening, or learn the new machine well enough to run it.

AI Is the Press



Run it, or someone else will.

YOU CAN START NOW

Here's the great news: you don't have to wait until college or your first job to get good at this. You can start now. Learn what AI does well, notice where it struggles, and practice using it for something you care about. Every project gives you experience you can bring to the next one.

Before personal computers, becoming a designer meant years at a drafting table learning design by hand. Then desktop publishing put powerful design tools on a teenager's desk. They could create posters, magazines, and brochures while developing their skills. The tool didn't replace skill. It shortened the distance between wanting to do the work and actually doing it.

Why You'll Thrive in the AI Future



This Is Your Time

Start building AI skills while you're in school. Try new things, ask questions, and learn from what happens.



You'll Move Faster

Try ideas and get feedback on your work. AI can help you tackle projects you wouldn't have known how to start. Learn by building, starting now.



Build Good Habits Early

Use AI to learn and work from the start. Practice asking good questions, checking answers, and making your own decisions. These habits will stay with you.

Start now. Build skills you'll carry into whatever comes next.

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THIS HAS HAPPENED BEFORE

AI isn't the first technology to transform how people work. The steam engine did it for physical labor, electricity for factories, and the internet for information. **AI could do it across almost everything.**

In July 2025, the White House released an official national strategy document, *Winning the Race: America's AI Action Plan*. Here's how it describes what AI makes possible:

> "An industrial revolution, an information revolution, and a renaissance—all at once. This is the potential that AI presents.
>
> — **WINNING THE RACE** · The White House, July 2025

AI is today's printing press.

Learn to run it.

START SMARTER

AI is Different

Today's AI behaves differently from traditional software. If you expect it to act like the software you're used to, its behavior can be confusing.

Two differences help explain why. First, AI learns patterns from examples rather than relying only on rules written by programmers. Second, it can work with "unstructured data," such as everyday language, images, and audio, with far greater flexibility than traditional software. These differences make AI more capable in many situations, but also less predictable.

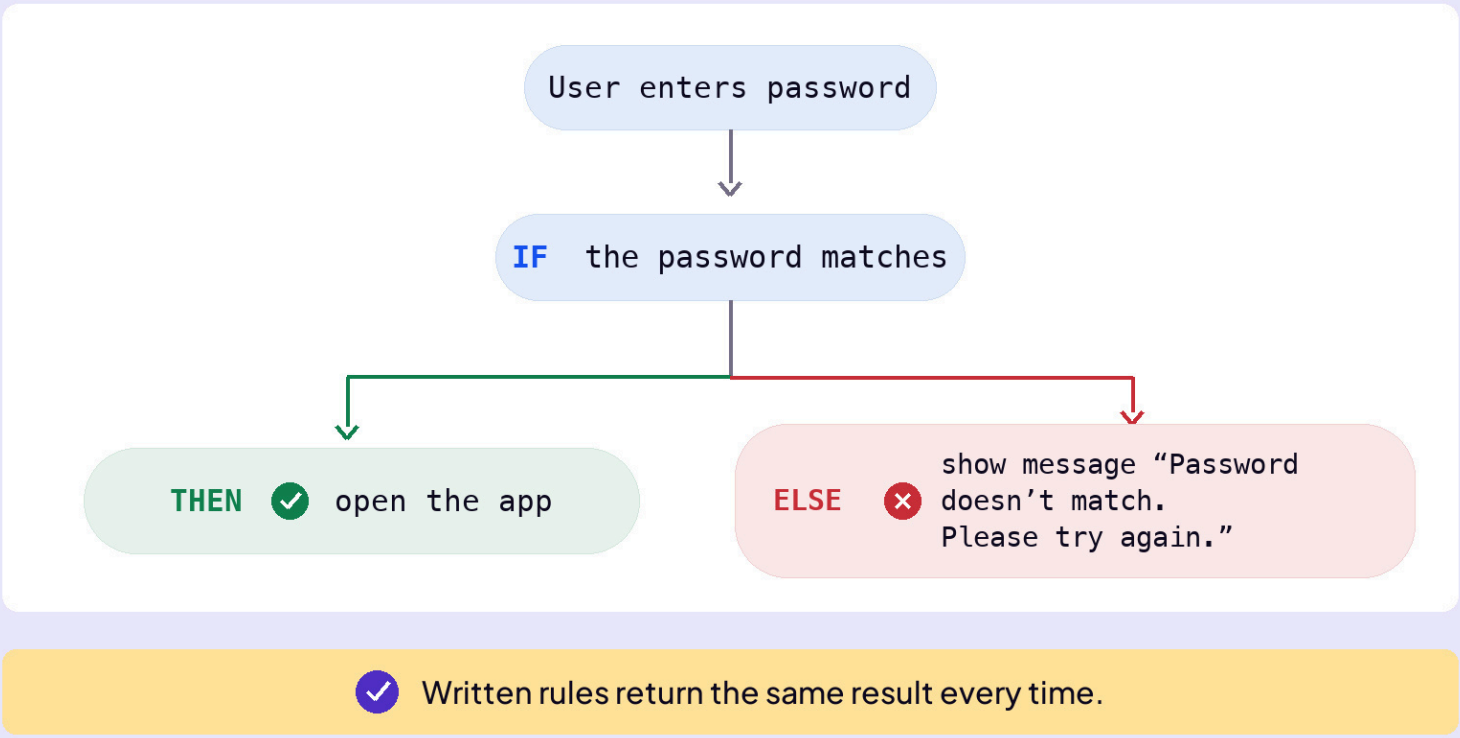
RULES-BASED SOFTWARE

You're signing into Spotify. You enter the wrong password. Should the app open because you were close? Or because your spelling was creative?

No. You want it to follow a rule: **let me in only if the password matches.**

Non-AI software is based on instructions like this that people wrote. A calculator follows instructions for mathematical operations. A spreadsheet applies formulas to the information you enter.

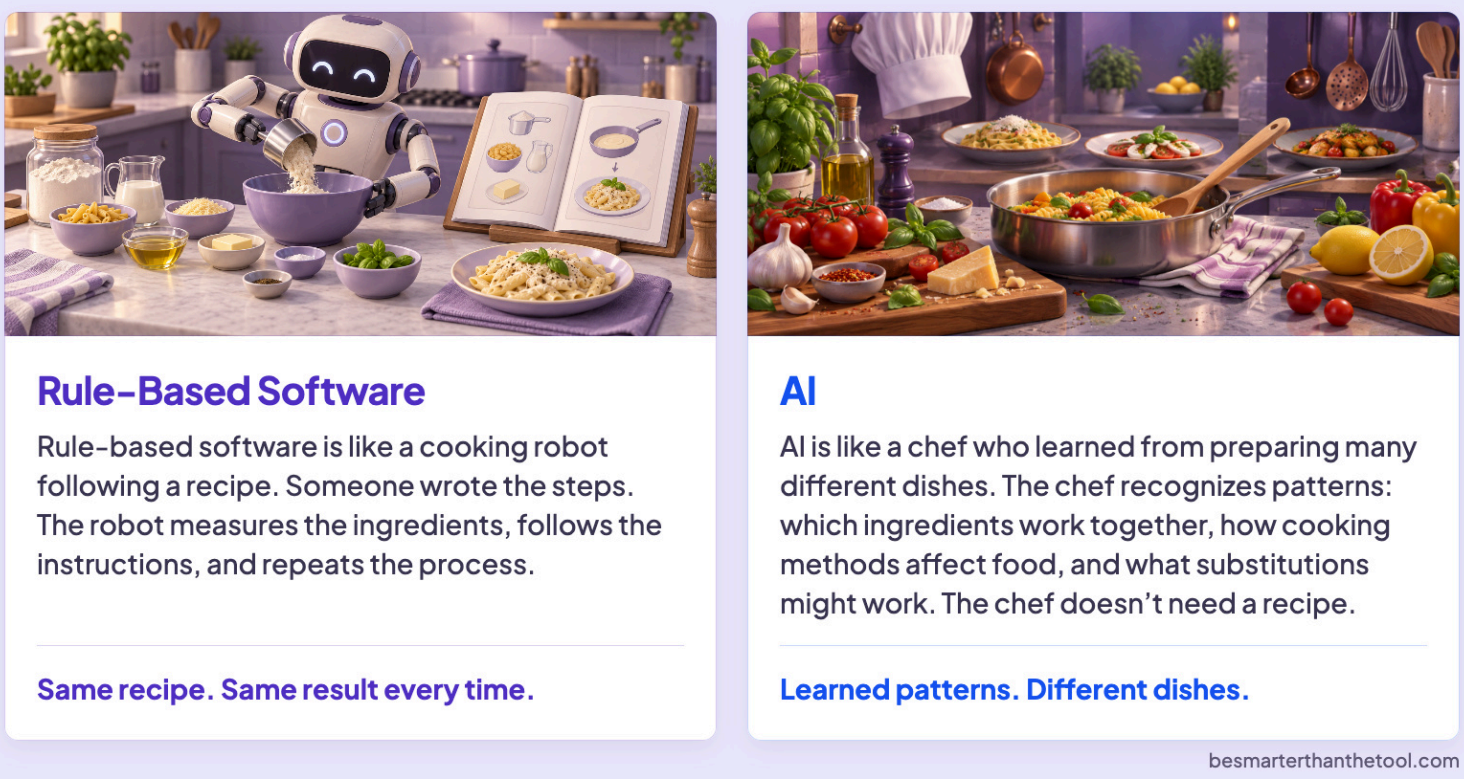
Rules Look Like This



RULES VS. PATTERNS

Traditional software follows instructions written by programmers. AI learns patterns from examples during training and uses those patterns to answer you. People still write the code that trains and runs AI, but they don't write each answer.

A Cooking Analogy



STRUCTURED VS. UNSTRUCTURED DATA

Traditional software usually needs information entered in a specific structure. Think of a spreadsheet, with rows and columns. That's **structured data**.

AI can work with **unstructured data**. You don't need to organize it into specific rows and columns. Give it what you have, and it can help make sense of it.

Here's the same short playlist in both formats. Notice the difference?

Same Playlist. Different Formats.

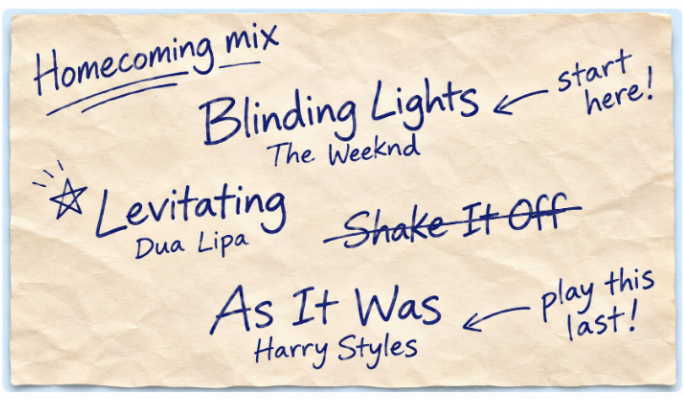
Traditional Software

Structured Data

Song	Artist	Play Order
Blinding Lights	The Weeknd	1
Levitating	Dua Lipa	2
As It Was	Harry Styles	3

AI

Unstructured Data



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AI can do this because, during training, it learned patterns in handwriting, language, and visual clues like arrows and crossed-out words. It uses those patterns to help interpret your notes, even when they aren't neatly organized.

THE DOWNSIDE

Using learned patterns makes AI flexible, but its response can be harder to predict and inspect than a written rule like the password check.

It might miss a crossed-out song. It might look great, but still have mistakes.

Different jobs need different tools.

Choose the tool that fits the work.

START SMARTER

What's an LLM?

If you've used AI, you've probably used an app like ChatGPT, Claude, or Gemini. The app is what you use. Under the hood, a **Large Language Model**, or **LLM** for short, does the work.

What's an LLM?



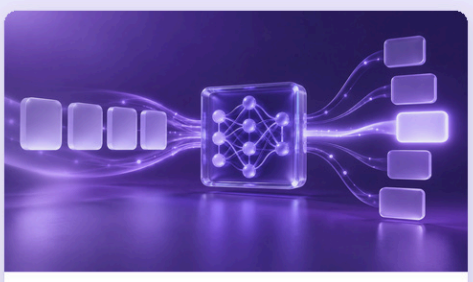
Large

Trained on huge amounts of text and code.



Language

Reads, writes, summarizes, translates, and explains.



Model

Predicts likely output from learned patterns.

ChatGPT is the app. The LLM is the engine.

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LEARNING PATTERNS

Before you use it, the model goes through training, learning patterns from enormous amounts of text and other data.

Here's one you picked up as a child. Which word comes next?

Patterns AI Learns



One Familiar Pattern

Peanut butter and ___ → **jelly**

You knew it. So does AI.



Patterns Are Everywhere

Twinkle, twinkle, little ___ → **star**

Once upon a ___ → **time**

Better late than ___ → **never**

AI learns patterns by working through billions of examples.

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These are easy patterns you already know. AI also learns patterns in how people explain ideas, ask questions, solve problems, and even misspell words.

BUILDING AN ANSWER

When you ask a question, the model uses those learned patterns and the words already there to work out what could come next. It chooses a likely next word, adds it, and repeats.

The words already there change what might come next. "Peanut butter and…" might lead to "jelly." "Peanut butter and banana…" might lead to "sandwich."

Your phone does something similar when you write a text: it suggests a word, you tap it, and it suggests the next. An LLM keeps going to build an answer.

One Word at a Time

I want to buy peanut butter and → **jelly**

I want to buy peanut butter and **jelly** → **for**

I want to buy peanut butter and jelly **for** → **lunch**

Add a word. Use the updated sentence. Predict again.

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Each added word becomes part of what the model uses for its next prediction. More than one word can fit, so the same starting text can lead to different answers.

Training builds the patterns.

The model uses those patterns to build your answer, one word at a time.

START SMARTER

Probability & Prediction

What's the magic that powers ChatGPT, Claude, and every other AI you've used? **Math**.

A big part of that math is probability: how likely something is. When AI builds an answer, it calculates probabilities for what comes next.

WHERE PROBABILITY MATH BEGAN

In 1654, two French mathematicians, Blaise Pascal and Pierre de Fermat, traded letters about gambling. Their work helped lay the foundation for modern probability.

Start with something simple: when every outcome is equally likely, you can calculate a probability by counting the possibilities.

Standard Probability

$$\frac{\text{Ways to get the result}}{\text{Total possible outcomes}} = \text{PROBABILITY}$$

Ways to get the result ÷ total possible outcomes = probability.

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Counting the Possibilities

THE SCENARIO

You toss two coins. What's the chance that both land on heads?



$$\frac{\text{Ways to get two heads}}{\text{Total possible outcomes}} = \frac{1}{4} = 25\%$$

Before new evidence: 1 out of 4 = 25%.

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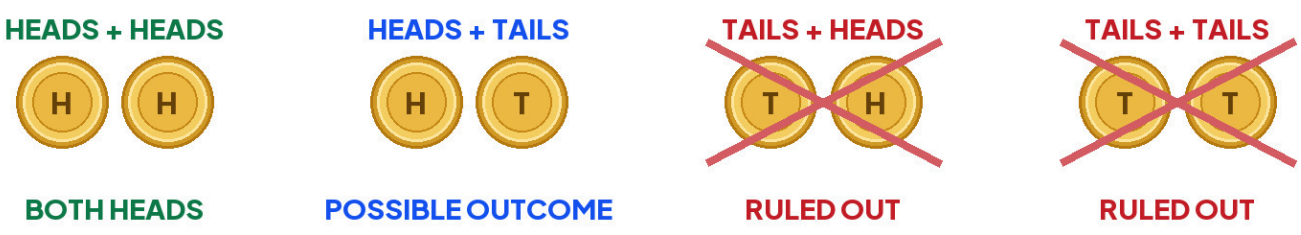
CONDITIONAL PROBABILITY

New evidence can change the odds. Conditional probability takes that evidence into account.

A Clue Changes the Odds

THE SCENARIO

You toss two coins. Someone peeks and tells you the first coin landed heads. What's the chance that both coins landed heads now?



$$\frac{\text{Ways to get two heads}}{\text{Total possible outcomes}} = \frac{1}{2} = 50\%$$

After the clue: 1 out of 2 = 50%.

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The coins didn't change when someone peeked. What you knew about them did. That changed the odds from **25%** to **50%**.

PREDICTING WHAT COMES NEXT

AI uses conditional probability to build answers. Your question and the words already written shape the chances of what comes next. Each new word joins that text, and the process repeats.

What Comes Next?

YOU

What should I name my new dog?

AI'S REPLY SO FAR

You could name him ____

A FEW POSSIBLE NEXT WORDS

Spot	Max	Buddy
22%	17%	14%

Illustrative probabilities. Other possible next words make up the remaining 47%.

The question and the words already written shape what is likely to come next.

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AI builds answers with probabilities.

One prediction at a time.

START SMARTER

Does AI Think?

You type a question and AI types back like a person. It explains, it jokes, it says sorry when it slips up. After a while it's hard not to feel like someone is in there. So here's the big question: **Does AI actually think?** When it answers like a person, is it understanding things the way you do?

SOUNDING HUMAN

AI can finish your sentence, explain a poem, and carry on a conversation. But sounding like a person doesn't tell you whether it understands things the way a person does.

There's a famous way to picture this, sometimes called the Chinese Room. Imagine someone locked in a room who doesn't know a word of Chinese. Notes in Chinese come in under the door. The person can't read any of it, but they have a giant rulebook telling them which symbols to send back. So they match symbols and slide back an answer that looks perfect to the people outside.

The Chinese Room



A convincing answer doesn't prove understanding.

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The Chinese Room illustrates a distinction: producing an answer and understanding it aren't necessarily the same thing. An LLM doesn't follow a literal rulebook. It uses patterns learned during training to predict its response.

REAL TEXT. WRONG MEANING.

This actually happened. When searchers asked Google "how do I keep cheese from sliding off my pizza," the AI suggested mixing about one-eighth of a cup of non-toxic glue into the sauce.

That bizarre answer did not come from nowhere. It traced back to an old joke on Reddit that was written like sincere advice. Google's AI found real text, but it missed what the text meant.

Real Text. Wrong Meaning.



The Reddit comment was real. The cooking advice was not.

The source was real, but AI interpreted it incorrectly.

When You Think. What AI Does.



When You Think	What AI Does
Experience and meaning. You draw on things you have lived through. You connect words to what they mean in your life.	AI uses learned patterns and what you tell it.
Word choice. You choose words to express what you want to say.	AI predicts the next word, then repeats.
Uncertainty. You can notice when you are unsure.	AI can sound certain even when it is wrong.

Similar-looking answers can come from very different processes.

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None of this means AI is dumb or useless. It can do impressive work. But an answer that sounds human is not enough to show that it understands the way you do.

Sounds human. Works differently.

A convincing answer doesn't prove understanding.

START SMARTER

Does School Matter?

A lot of people your age are quietly asking the same thing. If AI can answer anything, do you even need school anymore? Why sit through years of classes when the tool spits out the answer in seconds?

Think about it this way.

Fast forward a few years. You've landed your dream job, and AI sits right next to you all day, cranking out drafts, code, and plans. Your coworker at the next desk is doing the same. You each have the same job, using the same AI.

Here's the catch: ask AI similar questions, and you often get similar answers. They might be good. They might need work. Either way, that starting point is available to everyone. It's the new average. What you bring to the work is what takes you further.

Same Tool. Different Advantage.



The tool may be the same. What you bring to it is yours.

That leads to a question worth considering: In your AI future, what takes you beyond the new average? What makes you more valuable?

WHAT SETS YOU APART

Your secret sauce is the knowledge and skills you build. And where do you build them? You know where this is going: school.

School is where you get to build what you bring to the work. Learn how things work. Practice writing, solving problems, making things, and working with people. Find something that interests you and go deep. Whatever tools come next, that knowledge and those skills are yours. That's what it means to **Be Smarter Than the Tool.**

What School Helps You Build



Knowledge

Go deep in a subject you care about. Become someone people turn to for your knowledge.



Skills

Learn to write, build, solve problems, and work with people. And practice thinking critically. Question claims, weigh evidence, and make informed decisions.

School is where you get to build what you bring to the work.

The opportunity to learn is already in front of you.

Choose something you want to get better at. Start there.

START SMARTER

Learn with AI

Do you ever feel like there aren't enough hours in the day? You might have a history quiz tomorrow, an essay due Friday, and what feels like a never-ending stream of algebra homework. Add sports, activities, family, and everything else, and finding enough time to study feels impossible.

There's an easy solution: add four more hours to every day. Then you'd have plenty of time.

Of course, that isn't possible. AI can't add hours to your day, either.

THE PATIENT TUTOR

Here's the good news: AI might be the best study tool you'll ever get your hands on. Used the right way, it helps you finish the work and makes you genuinely sharper at the subject.

Believe it or not, this is one place where a machine has an advantage over a human tutor. AI will quiz you at 1 a.m., never sigh when you ask again, and meet you exactly where you're stuck.

The first move is choosing which kind of AI to aim at the job. There's a simple question to guide your choice: do you want to learn from materials you already have, or learn something new?

Which Study Tool for the Job?



GEMINI NOTEBOOK

Focus

Uses your materials to answer questions and create study tools. That keeps your exam prep focused on what you need to learn.

BEST USE

You have the materials the test covers, including class notes, screenshots, study guides, a webpage, or a YouTube video.

THE CATCH

If your notes are missing a key concept, it cannot reliably fill the gap.



CHATGPT · CLAUDE · GEMINI

Exploration

Makes things from scratch. Trained on massive amounts of data, it can explain or quiz you on almost anything.

BEST USE

You do not have the materials, or you want a concept explained a new way or extra practice your notes do not cover.

THE CATCH

It can make things up or explain a concept differently from how your teacher taught it and expects it on the exam.

Choose the tool that matches how you need to learn.

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WHY GEMINI NOTEBOOK

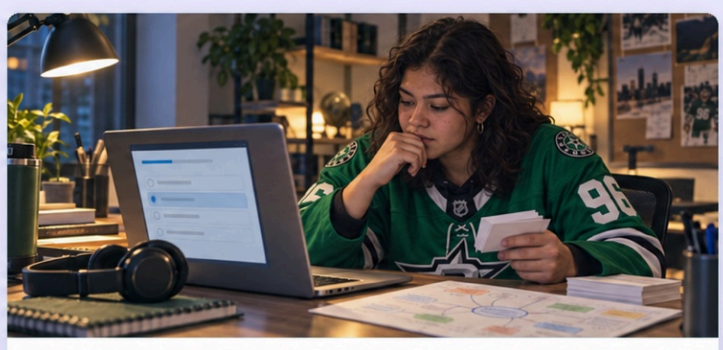
Use Gemini Notebook as your main study tool when you're studying from class materials. It keeps your work grounded in the notes, slides, and sources your teacher gave you.

How Gemini Notebook Works



You Upload

Your notes and PDFs, slides, images, websites and articles, YouTube videos, and more.



You Get

Quizzes, flashcards, study guides, audio and video overviews, mind maps, and more.

Turn your class materials into the study tools you need.

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Your Four Moves for Gemini Notebook



One Subject per Notebook

Mixing subjects is like asking, "Quiz me on history, but make some of the questions about math."



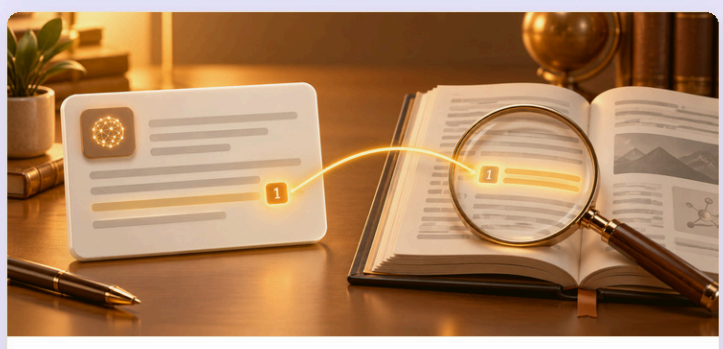
Give It the Full Picture

Add your notes, slides, videos, and anything else you need to learn.



Quiz Yourself Blind

Tell Gemini Notebook: "Generate a 10-question quiz from these sources only. Don't show the answers yet. Wait for mine, then grade me."



Trace It Back

If something is unclear, click the citation. It takes you back to the original material.

Use AI to strengthen the learning, not skip it.

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Use AI to learn, not to skip the learning.

Feed it your materials. Trace it back.